

Some Results on Continuous Functions

Definition 1 : Let $f: (a, b) \rightarrow \mathbb{R}$ be a function. If f is continuous at a point $x \in (a, b)$, then we have

$$\lim_{t \rightarrow x} f(t) = f(x+) = f(x-) = f(x).$$

Let $E_f = E = \{x \in (a, b) : f \text{ is discontinuous at } x\}$. Then we say that

- (1) f has a *discontinuity of the first kind* (or *simple discontinuity*) at x if both left $f(x-)$ and right $f(x+)$ limits exist and either $f(x+) \neq f(x-)$ or $f(x-) = f(x+) \neq f(x)$. In the former case, we say that f has a *jump discontinuity* at x , while in the latter case, we say that f has a *removable discontinuity* at x .
- (2) f has a *discontinuity of the second kind* if at least one of the limits $f(x-)$ or $f(x+)$ do not exist.

Observations/Examples :

1. Every linear function of the form $f(t) = at + b$, where $a, b \in \mathbb{R}$ is continuous. In fact, a polynomial function $p(t) = a_0 + a_1t + a_2t^2 + \dots + a_nt^n$, with $a_i \in \mathbb{R}$ ($1 \leq i \leq n$) is continuous.
2. (Dirichlet Function) Consider the function f on $\mathbb{R}$ given by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Then the Dirichlet function f has discontinuity of the second kind at each $x \in \mathbb{R}$. (Solution : Let $x \in \mathbb{R}$. If (t_n) is a sequence of rationals in either $(-\infty, x)$ or (x, ∞) with $\lim_{n \rightarrow \infty} t_n = x$, then we see that $(f(t_n))$ is the constant sequence with all terms 1 converging to 1. On the other hand, if (t_n) is a sequence of irrationals with $\lim_{n \rightarrow \infty} t_n = x$, then we see that $(f(t_n))$ is the constant sequence with all terms 0 converging to 0. Hence, neither $f(x-)$ nor $f(x+)$ exist.)

3. The function $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}, \end{cases}$ is continuous only at the point 0. (Solution : We see that $|f(x) - f(0)| = |f(x)| \leq |x|$ for all $x \in \mathbb{R}$. This shows that $\lim_{t \rightarrow 0} f(t) = 0 = f(0)$ and hence, f is continuous at 0. If $x \in \mathbb{R} \setminus \{0\}$, then as in the last example take a sequence (t_n) either in $(-\infty, x)$ or (x, ∞) with all rationals or with all irrational terms converging to x . If (t_n) is a sequence of rationals, then the sequence $(f(t_n)) = (t_n)$ converges to $x \neq 0$. On the other hand, if (t_n) is a sequence of irrationals, then the sequence $(f(t_n))$ with all terms 0 converges to 0. Thus neither $f(x-)$ nor $f(x+)$ exist at $x \neq 0$.)

Definition 2 : A function $f: (a, b) \rightarrow \mathbb{R}$ is said to be *monotonically increasing* (or *monotonically strictly increasing*) if $a < x < y < b$ implies that $f(x) \leq f(y)$ (resp. $f(x) < f(y)$). Similarly, f is said to be *monotonically decreasing* (or *monotonically strictly decreasing*) if $a < x < y < b$ implies that $f(x) \geq f(y)$ (resp. $f(x) > f(y)$).

By a monotonic function, we mean either a monotonically (strictly) increasing or monotonically (strictly) decreasing function.

Theorem 1 : For a monotonically increasing function $f: (a, b) \rightarrow \mathbb{R}$ and $x \in (a, b)$, we have

$$f(x-) = \sup\{f(t) : t \in (a, x)\} \leq f(x) \leq f(x+) = \inf\{f(t) : t \in (x, b)\}.$$

Similarly, for a monotonically decreasing function $f: (a, b) \rightarrow \mathbb{R}$ and $x \in (a, b)$, we have

$$f(x-) = \inf\{f(t) : t \in (a, x)\} \geq f(x) \geq f(x+) = \sup\{f(t) : t \in (x, b)\}.$$

Thus we conclude that a discontinuous monotonic function has only jump discontinuity. (Proof : Assume f is monotonically decreasing. (If f is monotonically increasing, then replacing f with $-f$, we can prove the stated result.) Consider the subset $\{f(t) : t \in (a, x)\} \subseteq \mathbb{R}$, which is bounded below by $f(x)$. Thus $\inf\{f(t) : t \in (a, x)\} = A$ exists and $A \geq f(x)$. We shall show that $A = f(x-)$. For a given $\epsilon > 0$, we see that $A + \epsilon$ is not an infimum of $\{f(t) : t \in (a, x)\}$. Thus $\exists t_0 \in (a, x)$ such that $A \leq f(t_0) < A + \epsilon$. Now for $t \in (t_0, x)$, as f is monotonically decreasing, we have $f(t_0) \geq f(t)$. Taking $\delta = x - t_0 > 0$, we see that for $t \in (x - \delta, x)$, we have $A \leq f(t) < A + \epsilon$. Hence, $A = f(x-)$. Similarly, we see that $\sup\{f(t) : t \in (x, b)\} = B$ exists and $f(x) \geq B$. Again for a given $\epsilon > 0$, we see that $B - \epsilon$ is not a supremum of $\{f(t) : t \in (x, b)\}$. Thus $\exists t_1 \in (x, b)$ such that $B \geq f(t_1) > B - \epsilon$. Now for $t \in (x, t_1)$, as f is monotonically decreasing, we have $f(t) \geq f(t_1)$. Taking $\delta = t_1 - x > 0$, we see that for $t \in (x, x + \delta)$, we have $B - \epsilon < f(t) \leq B$. Hence, $B = f(x+)$.)

Observations / Examples :

1. Consider the function $f(x) = \lfloor x \rfloor$ for $x \in \mathbb{R}$, where the *floor* $\lfloor x \rfloor$ of x is defined as the greatest integer $\leq x$. Clearly, the floor function is monotonically increasing. For an integer $n \in \mathbb{Z}$, we see that $f(n-) = n - 1$ and $f(n+) = n = f(n)$. Thus f has a jump discontinuity at each integer. For $x \in \mathbb{R} \setminus \mathbb{Z}$, we have an integer n such that $n < x < n + 1$. Further $f(x) = n$ for all $x \in (n, n + 1)$. Thus the floor function f is continuous at all non integer points. In other words, the set E_f of discontinuities of f is $\mathbb{Z}$.
2. If f is monotonically decreasing and $a < x < y < b$, then $f(x+) \geq f(y-)$. (Proof : From Theorem 1, we have $f(x+) = \sup\{f(t) : t \in (x, b)\} \geq \sup\{f(t) : t \in (x, y)\}$ and $f(y-) = \inf\{f(t) : t \in (a, y)\} \leq \inf\{f(t) : t \in (x, y)\}$. Since $\inf\{f(t) : t \in (x, y)\} \leq \sup\{f(t) : t \in (x, y)\}$, we see that $f(x+) \geq f(y-)$).

Theorem 2 : A monotonic function $f: (a, b) \rightarrow \mathbb{R}$ has countably many jump discontinuities. (Proof : Let E be the set of discontinuities of f . Assume that f is monotonically decreasing and from Theorem 1 all its discontinuities are jump discontinuities. (Replace f by $-f$ if f is monotonically increasing.) If $x \in E$, then $f(x-) > f(x+)$. Choose a rational number $r_x \in \mathbb{Q}$ such that $f(x-) > r_x > f(x+)$. Consider the map $x \mapsto r_x$ from E to $\mathbb{Q}$. Claim : The map $x \mapsto r_x$ is injective. If $x, y \in E$ and $a < x < y < b$, then we have $f(x-) > r_x > f(x+) \geq f(y-) > r_y > f(y+)$. Hence $x \neq y$ in E implies that $r_x \neq r_y$. As $\mathbb{Q}$ is countable, so is E .)

The converse of Theorem 2 holds. Given a countable (infinite) subset E of (a, b) , It is possible to construct a monotonic function f such that the set E_f of discontinuities of f is precisely E . First suppose that E is bounded below and $(a, b) \setminus E$ is a countable union of disjoint open intervals. Then we can express $(a, b) = \coprod J_n = J_1 \coprod J_2 \coprod J_3 \coprod \dots$, where J_m 's are open interval with $\sup J_i = \inf J_{i+1}$. Choose an increasing real sequence $\alpha_1 < \alpha_2 < \dots < \alpha_n < \dots$. Now we define a function $f: (a, b) \rightarrow \mathbb{R}$ by $f(t) = \alpha_n$ for every $t \in J_n$. If $p \in E$, then $p = \inf J_m$ for a unique m and we define $f(p) = \alpha_m$. It is a bit surprising to see that even for a countable dense subset E , one can construct such a monotonic function f .

Definition 3 : Let A be a subset of a metric space (X, d) .

- (i) We say that A is *dense* in X if the closure $\overline{A} = X$.
- (ii) X is said to be a *separable metric space* if it has a countable dense set. (For example, the n -dimensional Euclidean space $\mathbb{R}^n$ is separable. In fact, $\mathbb{Q}^n$ is a countable dense subset of $\mathbb{R}^n$.)
- (iii) A subset B of X is said to be *nowhere dense* in X if the closure $\overline{B}$ of B contains no open ball. (A finite subset of $\mathbb{R}$ is nowhere dense. The subset $\{\frac{1}{n} : n \in \mathbb{N}\}$ is also nowhere dense in the real line. Is Cantor's ternary set nowhere dense? Yes. But the subset of rationals $\mathbb{Q}$ being dense in the real line is not nowhere dense. We can see that a closed subset F of X is nowhere dense in X if and only if the open set $X \setminus F$ is dense in X .)

Theorem 4 : Let E be a countable dense subset of (a, b) . Then there is a monotonically (strictly) increasing function $f: (a, b) \rightarrow \mathbb{R}$ such that $E_f = E$. (Proof: Let $E = \{p_1, p_2, p_3, \dots\}$. For $x \in (a, b)$, set $I_x = \{m \in \mathbb{N} : p_m < x\}$. Consider a convergent series $\sum_{n=1}^{\infty} c_n$ of positive terms. (For example, we could have taken $c_n = \frac{1}{2^n}$.) Now for every subset $I \subseteq \mathbb{N}$, we see that the sum $\sum_{n \in I} c_n$ is finite. Define a function $f: (a, b) \rightarrow \mathbb{R}$ by

$$f(x) = \sum_{n \in I_x} c_n ,$$

where summation runs over all the indices n such that $p_n < x$.

Step-1: f is strictly increasing - If $x, y \in (a, b)$ with $x < y$, then $I_x \subseteq I_y$. As E is dense in (a, b) , there exists $p_i \in (x, y) \cap E$, i.e., $i \in I_y \setminus I_x$. This shows that

$$f(x) = \sum_{n \in I_x} c_n < \sum_{n \in I_y} c_n = f(y).$$

Step-2: f is continuous at every $x \in (a, b) \setminus E$ - Let $\epsilon > 0$ be given. Since the series $\sum_{n=1}^{\infty} c_n$

is convergent, there exists $n_0 \in \mathbb{N}$ such that $\sum_{n=n_0+1}^{\infty} c_n < \epsilon$.

Let $\delta = \min\{|x - p_j| : 1 \leq j \leq n_0\} > 0$. Then $(x - \delta, x + \delta) \cap E \subseteq \{p_n : n > n_0\}$. For $s, t \in (x - \delta, x + \delta)$ with $s < x < t$, we see that $I_t \setminus I_s \subseteq \{n \in \mathbb{N} : n > n_0\}$. Thus, we have $0 < f(t) - f(s) = \sum_{n \in I_t} c_n - \sum_{n \in I_s} c_n \leq \sum_{n=n_0+1}^{\infty} c_n < \epsilon$. This shows that

$$\begin{aligned} 0 \leq f(x+) - f(x-) &= \inf\{f(t) : t \in (x, x + \delta)\} - \sup\{f(s) : s \in (x - \delta, x)\} \\ &\leq \inf\{f(t) - f(s) : x - \delta < s < x < t < x + \delta\} \\ &< \epsilon. \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, we must have $f(x+) = f(x-)$. Hence, f is continuous at every $x \in (a, b) \setminus E$.

Step-3 : f has a jump discontinuity at every $x \in E$ - Let $x = p_m \in E$. For a given $\epsilon > 0$, choose $n_1 \in \mathbb{N}$ such that $\sum_{n=n_1+1}^{\infty} c_n < \epsilon$, where $n_1 > m$. (We could have taken $n_1 = \max\{n_0, m + 1\}$.)

Let $\delta = \min\{|p_m - p_j| : 1 \leq j \leq n_1 \text{ and } j \neq m\} > 0$. Then

$$(x - \delta, x + \delta) \cap E \subseteq \{p_m\} \cup \{p_n : n > n_1\}.$$

For $s, t \in (x - \delta, x + \delta)$ with $s < x < t$, we have $m \in I_t \setminus I_s \subseteq \{m\} \cup \{n \in \mathbb{N} : n > n_1\}$. Thus

$$c_m < f(t) - f(s) \leq c_m + \sum_{n=n_1+1}^{\infty} c_n < c_m + \epsilon.$$

Now proceeding as in Step-2, we get $c_m \leq f(x+) - f(x-) \leq c_m + \epsilon$. Since ϵ is arbitrary, we have $f(x+) - f(x-) = c_m > 0$. Thus f has a jump discontinuity at every $x = p_m \in E$.

There is a (non-monotonic) function on $\mathbb{R}$ that is discontinuous precisely at rational numbers. One such function is *Thomae function* or *modified Dirichlet function*. The Thomae function has removable discontinuity at each rational point and it is given by

$$g(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \in \mathbb{Q} \text{ (} x \text{ in standard form),} \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Theorem 5 : The Thomae function is discontinuous precisely at rational points. Further, discontinuities of the Thomae function are all removable. (Proof : Let $x \in \mathbb{R} \setminus \mathbb{Q}$ and $\epsilon > 0$ be given. Choose $m \in \mathbb{N}$ such that $\frac{1}{m} < \epsilon$. Note that there are only finitely many rational numbers of the form $\frac{a}{b}$ with $1 \leq b \leq m$ lying in the open interval $(x - 1, x + 1)$. Let $\delta = \min\{|x - r| : r = \frac{a}{b} \in (x - 1, x + 1) \cap \mathbb{Q} \text{ with } 1 \leq b \leq m\}$. As x is irrational, $\delta > 0$. Let $t \in (x - \delta, x + \delta)$. If t is irrational, then $g(t) = 0$. If $t = \frac{a}{b}$ is rational, then $b > m$ and this shows that $g(t) = \frac{1}{b} < \frac{1}{m} < \epsilon$. Thus for every $t \in (x - \delta, x + \delta)$, we have

$$|g(t) - g(x)| = |g(t) - 0| < \epsilon.$$

Hence, $\lim_{t \rightarrow x} g(t) = 0 = g(x)$ and g is continuous at $x \in \mathbb{R} \setminus \mathbb{Q}$.

Let $x = \frac{p}{q} \in \mathbb{Q}$. Then $g(x) = \frac{1}{q} > 0$. Let $\epsilon > 0$ be given. Choose $m \in \mathbb{N}$ such that $\frac{1}{m} < \epsilon$. Let $\delta = \frac{1}{mq} > 0$. If $\frac{a}{b}$ is a rational number with $0 < |\frac{a}{b} - x| < \delta$, then we have

$$\frac{1}{mq} = \delta > \left| \frac{a}{b} - \frac{p}{q} \right| = \frac{|aq - bp|}{bq} \geq \frac{1}{bq},$$

as $|aq - bp| \geq 1$. Thus $b > m$. In other words, all rational numbers in the punctured open interval $(x - \delta, x + \delta) \setminus \{x\}$ has denominator $> m$. Now $t \in (x - \delta, x + \delta) \setminus \{x\}$ implies that $g(t) = 0$ if t is irrational and

$$g(t) = \frac{1}{b} < \frac{1}{m} < \epsilon,$$

if $t = \frac{a}{b}$ is rational. This shows that $\lim_{t \rightarrow x} g(t) = 0 \neq \frac{1}{q} = g(x)$. Hence, g has a removable discontinuity at each rational point.)